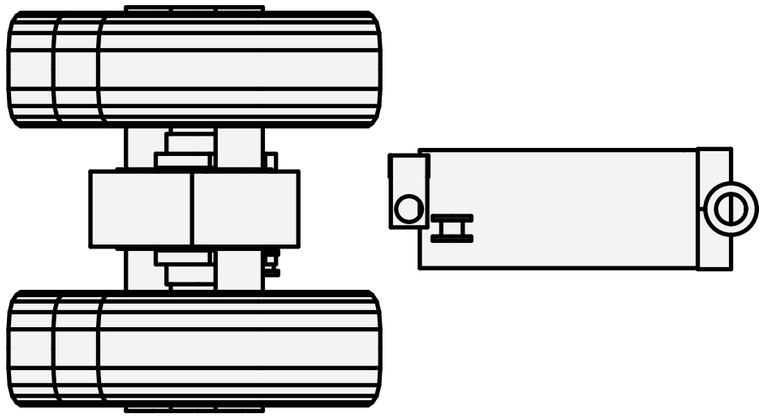
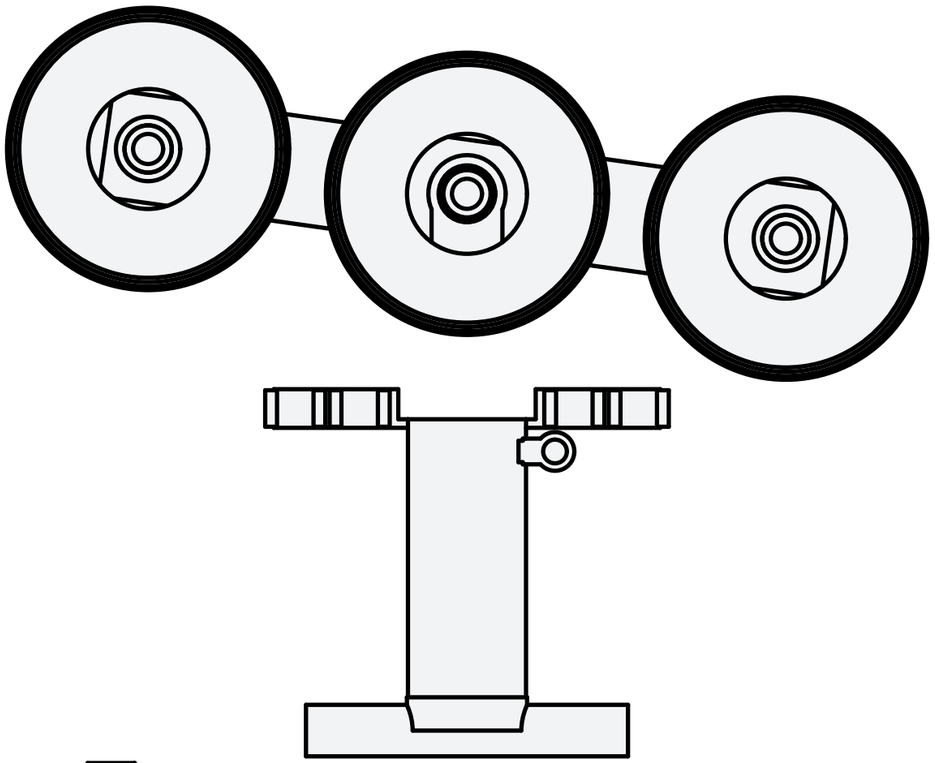
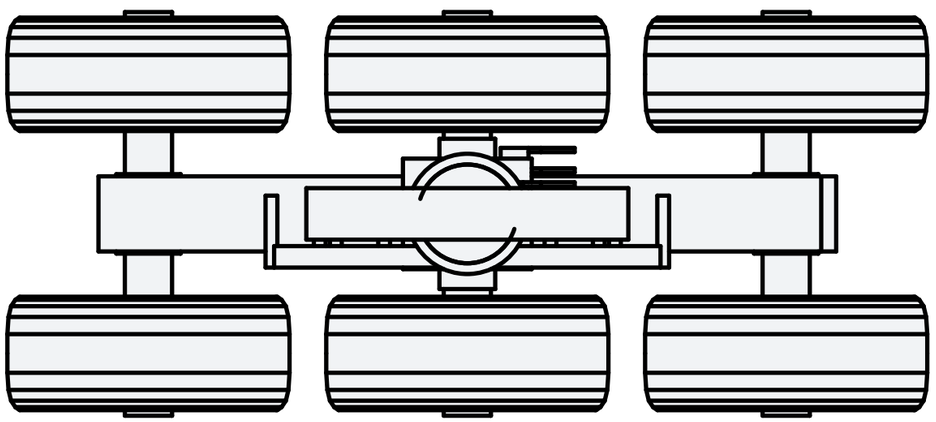
Overall landing gear assembly - orthographic arrangement

State S0 | Theta=0 deg | Stroke=0 mm | Beta=-8 deg | DoorAngle=180 deg | Downlock=1.0 Uplock=0.0



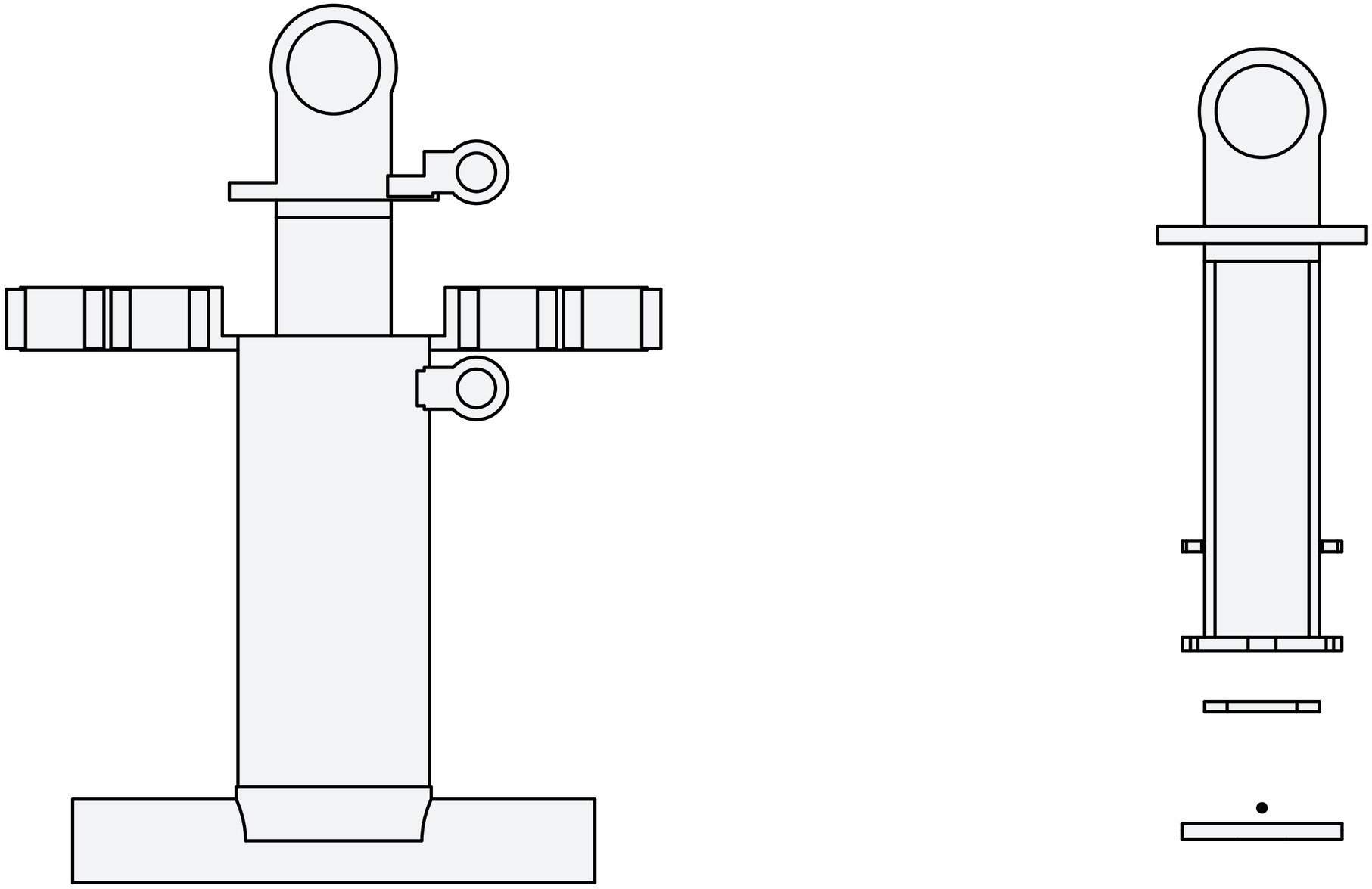




T1/T2 centres: x=-600 / +600 mm; leg datum B=(0, 0, -2500 + Stroke)
Bogie L/W/H: 3400 mm/350 mm/500 mm; axle pitch 1500 mm
Wheel centres y=+/-650 mm; tyre nominal R/W 657.225 mm/528.32 mm; overall view scale is provisional

Oleo longitudinal section - pressure internals and stops

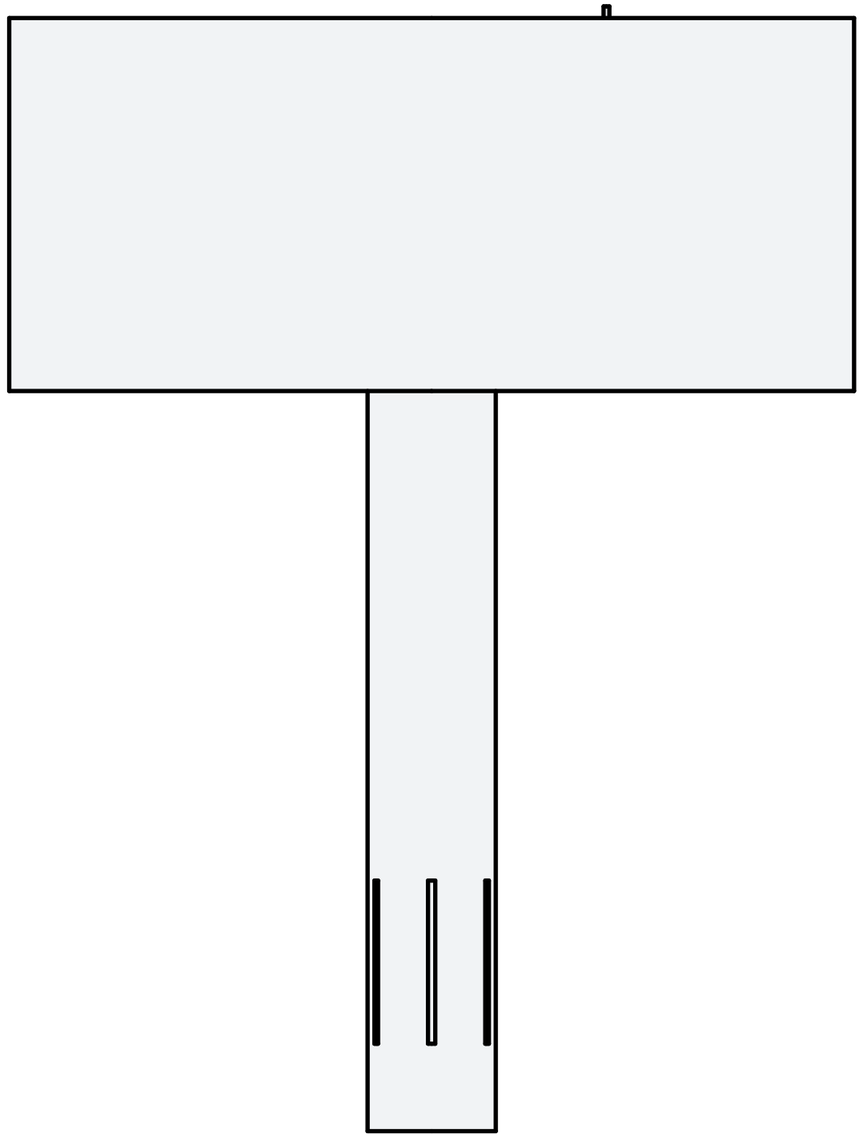
State S2 | Theta=0 deg | Stroke=279.379 mm | Beta=0 deg | DoorAngle=180 deg | Downlock=1.0 Uplock=0.0

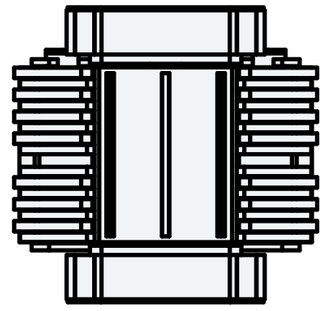


Barrel OD/ID: 550 mm/460 mm; barrel length 1450 mm
Piston OD/ID: 330 mm/270 mm; stroke max 600 mm; gas reference 65 L
Metering: 8 x dia 21.484 mm compression; 8 extra x dia 19.625 mm return; areas 2900/5320 mm2; section Y=0

FI wheel and front axle - bearing, brake and retention section

State S1 | Theta=0 deg | Stroke=120.666 mm | Beta=0 deg | DoorAngle=180 deg | Downlock=1.0 Uplock=0.0

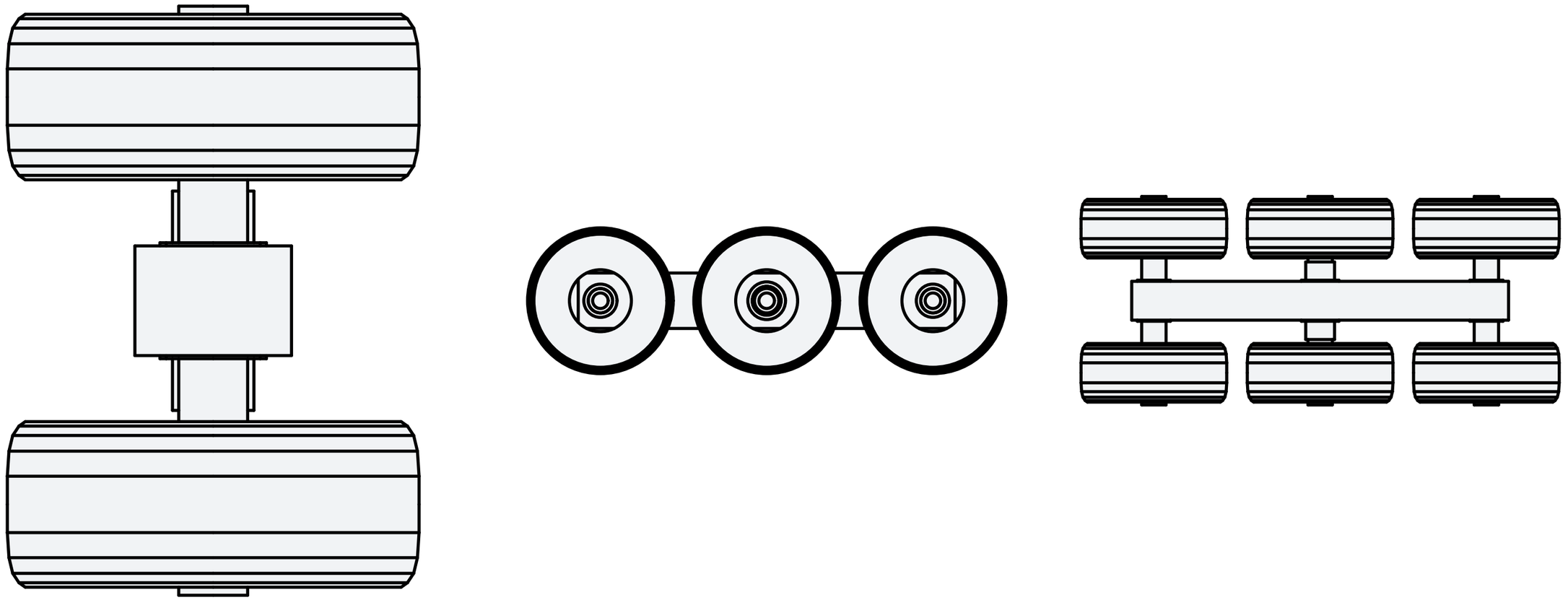




FI tyre nominal R/W: 657.225 mm/528.32 mm; rim R 279.4 mm
Front axle OD/ID: 220 mm/140 mm; bearing and seal stack shown as native FI PartIDs
Brake: 8 rotors OD/ID 520/290 x 18 mm; 8 stators 500/270 x 16 mm; section X=1500 mm

Bogie beam, three axles and six-wheel layout

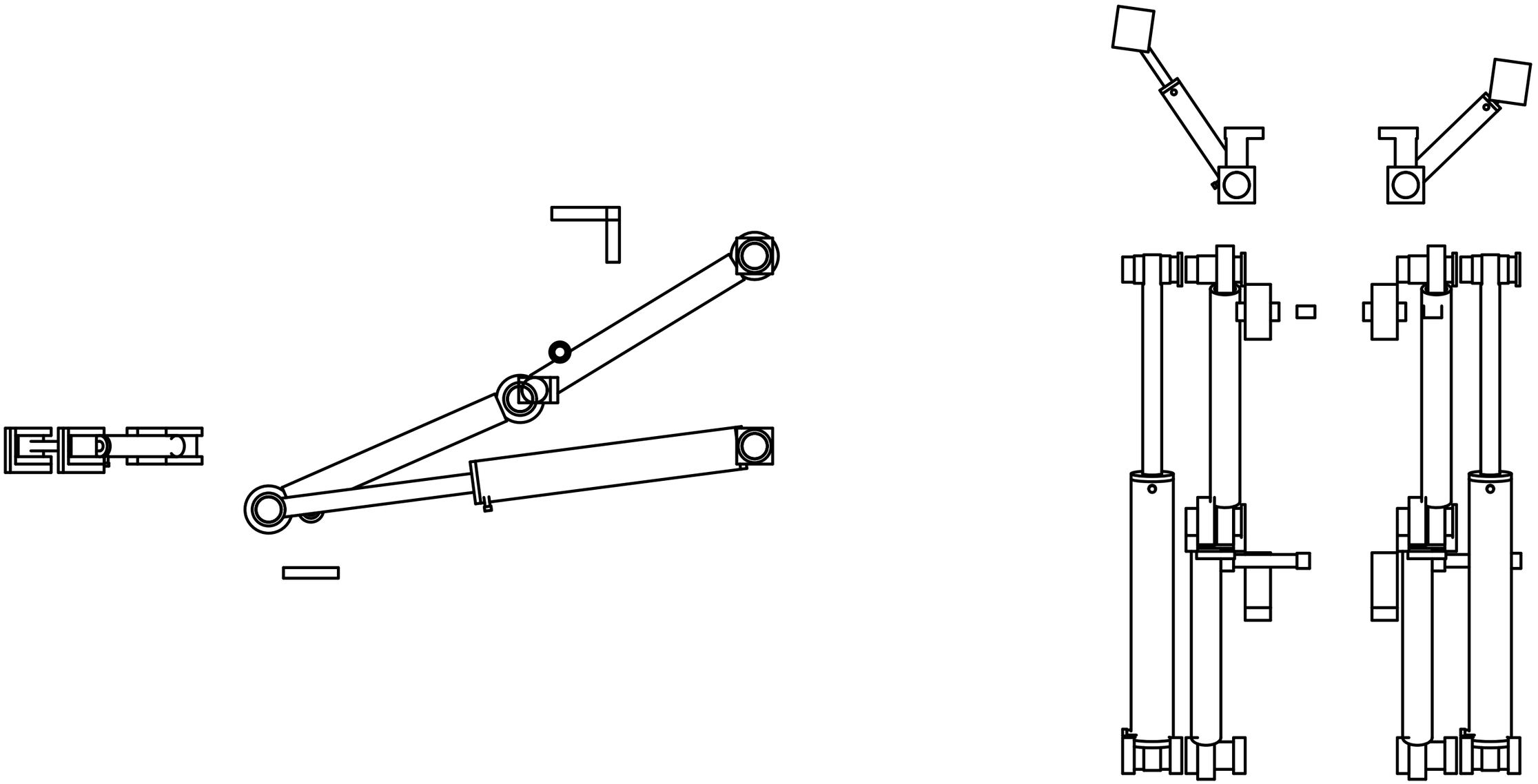
State S1 | Theta=0 deg | Stroke=120.666 mm | Beta=0 deg | DoorAngle=180 deg | Downlock=1.0 Uplock=0.0



Bogie L/W/H: 3400 mm/350 mm/500 mm; wall 35 mm
Axle pitch 1500 mm; centre axle OD/ID 260 mm/160 mm
Pivot pin OD 260 mm; wheel stations x=-1500/0/+1500 mm; bearing fits provisional

Folding stay, fixed-length actuators and mechanical locks

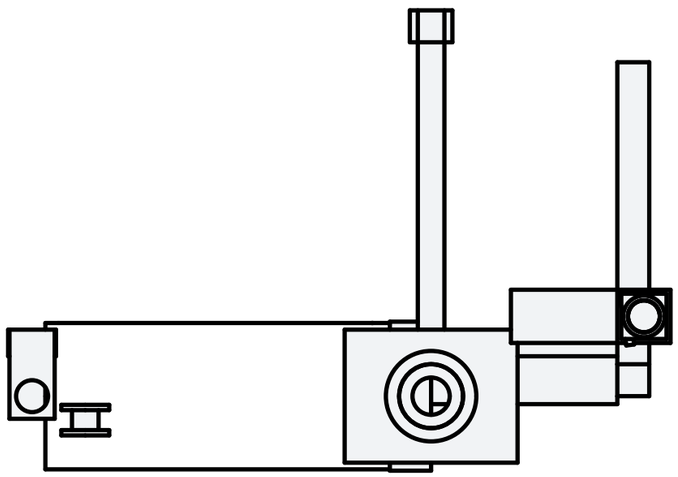
State S0 | Theta=0 deg | Stroke=0 mm | Beta=-8 deg | DoorAngle=180 deg | Downlock=1.0 Uplock=0.0

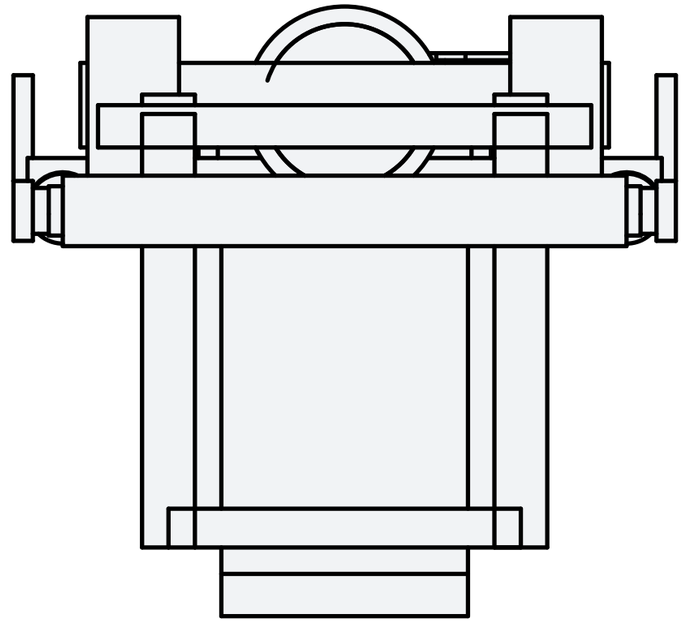


Stay links L1/L2=1300 mm fixed; knee positive pin withdraws in X before folding
Main actuator: bore 160 / rod 90 mm; body 1240 mm + fixed rod 1280 mm; telescopic overlap is material
Pitch actuator: bore 70 / rod 45 mm; body 505 mm + fixed rod 570 mm; clevis pins use 0.2 mm radial fit
Downlock pin, C-hook uplock and release rods are projected from native mechanism PartIDs
Kinematics source: RLG003_MechKinematics; pin/bore fits and door gaps are preliminary drawing values

T1/T2 trunnion bearings and rigid fixture interface

State S0 | Theta=0 deg | Stroke=0 mm | Beta=-8 deg | DoorAngle=180 deg | Downlock=1.0 Uplock=0.0

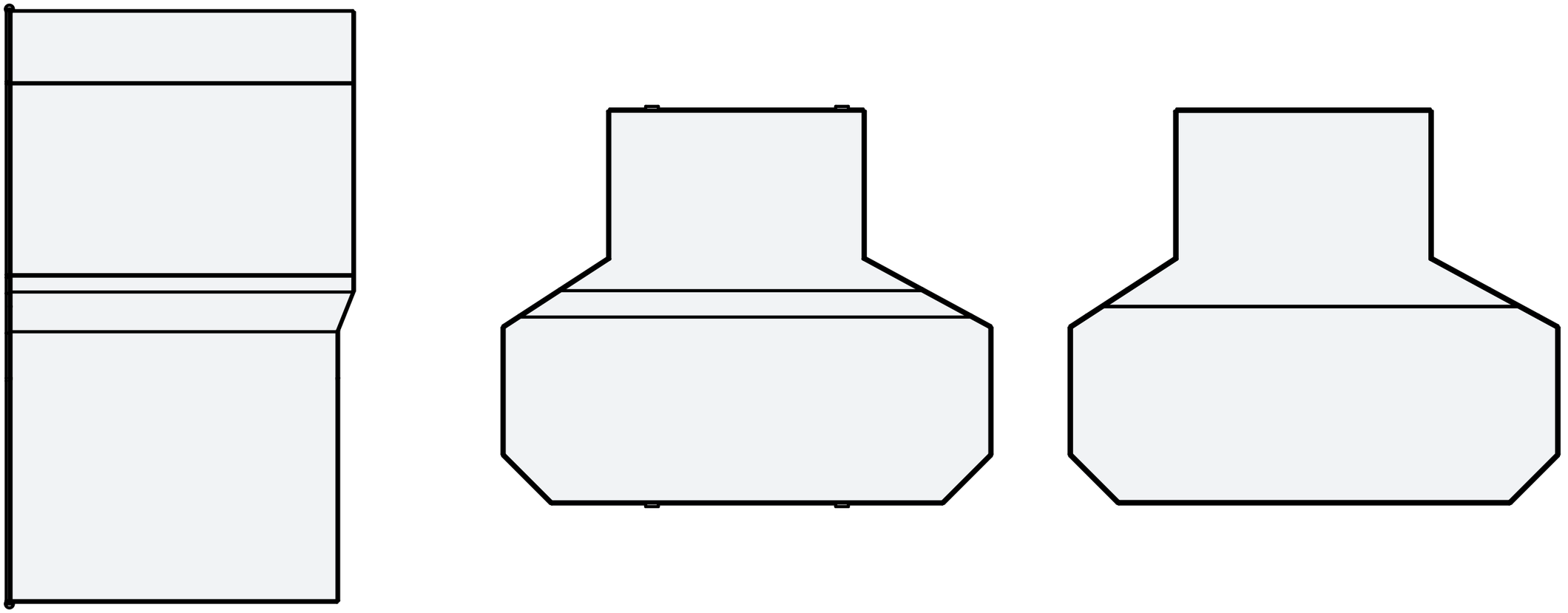




T1/T2 trunnion OD 240 mm; bearing sleeve ID nominal OD + 0.3 mm fit
Fixture towers, crossbar and actuator root frames are connected rigid supports in root coordinates
Stay crossbar y=-1255 mm; tie beams run toward y=-25 mm; fixture mass excluded from gear BOM
Kinematics source: RLG003_MechKinematics; pin/bore fits and door gaps are preliminary drawing values

Bay inner dimensions, opening rim and closed door section

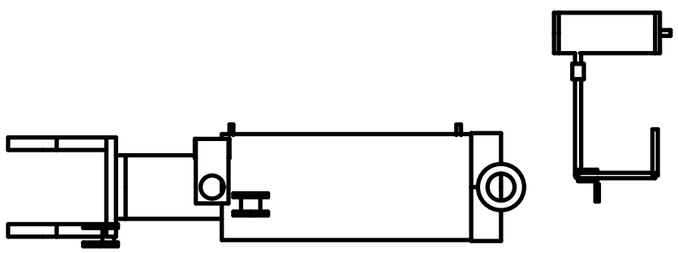
State S9 | Theta=90 deg | Stroke=0 mm | Beta=0 deg | DoorAngle=0 deg | Downlock=0.0 Uplock=1.0

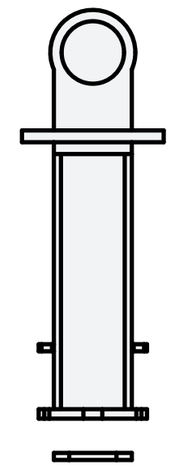


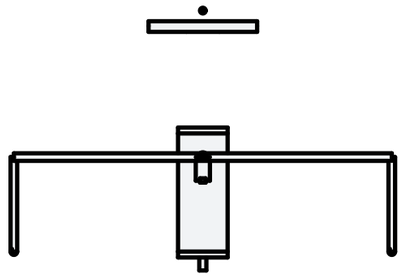
Bay inner envelope X=-2300..+2300, Y=-3400..+300, Z=-1100..+1100 mm
Closed doors: 10 mm plate; 2 mm perimeter gap, 4 mm centre seam; hinge centre Z=-1110 mm
Horizontal section Z=-1090 mm checks tyre clearance; doors open downward/outboard through 180 deg; stops stay outside tyre grown X

Conventional oleo internals and 12 L emergency accumulator details

State S1 | Theta=0 deg | Stroke=120.666 mm | Beta=0 deg | DoorAngle=180 deg | Downlock=1.0 Uplock=0.0



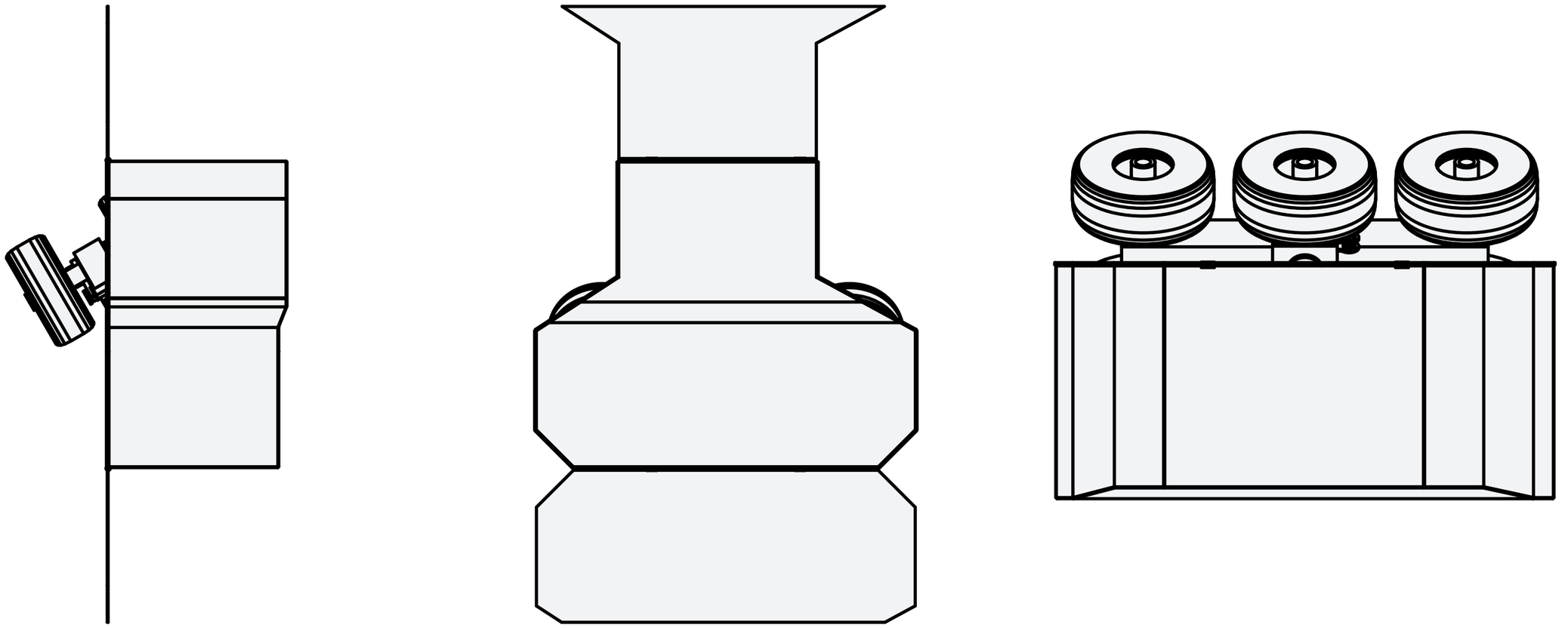




Conventional central oleo barrel, piston, metering and stop stack shown at section plane Y=0 mm
Accumulator shell: 12.0 L internal; gas/oil bladder boundary; precharge 10.0 MPa abs; EPDM membrane
Cam/bypass opens for Theta <= 20 deg; drilled charge/oil ports, isolation valve and connected lines are real solids

Motion state S6 - mid-fold with doors open

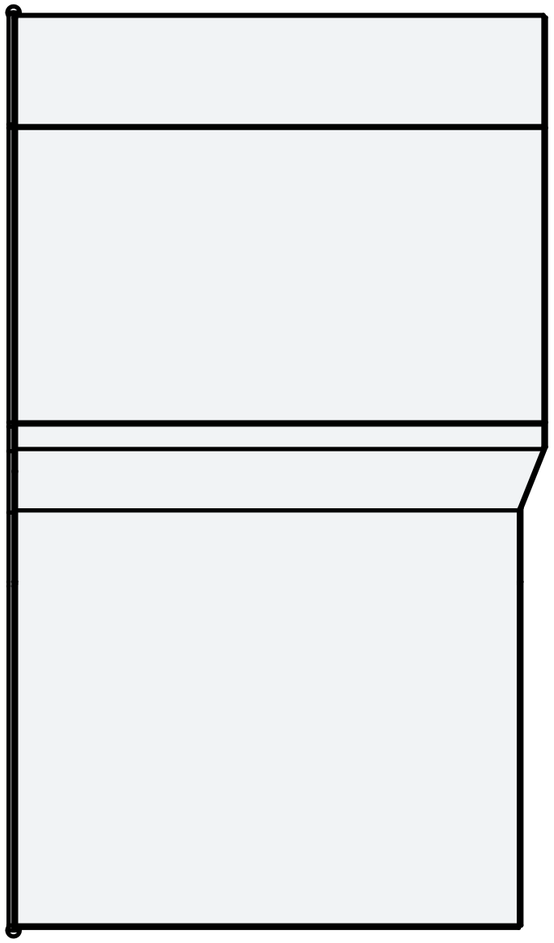
State S6 | Theta=60 deg | Stroke=0 mm | Beta=0 deg | DoorAngle=180 deg | Downlock=0.0 Uplock=0.0

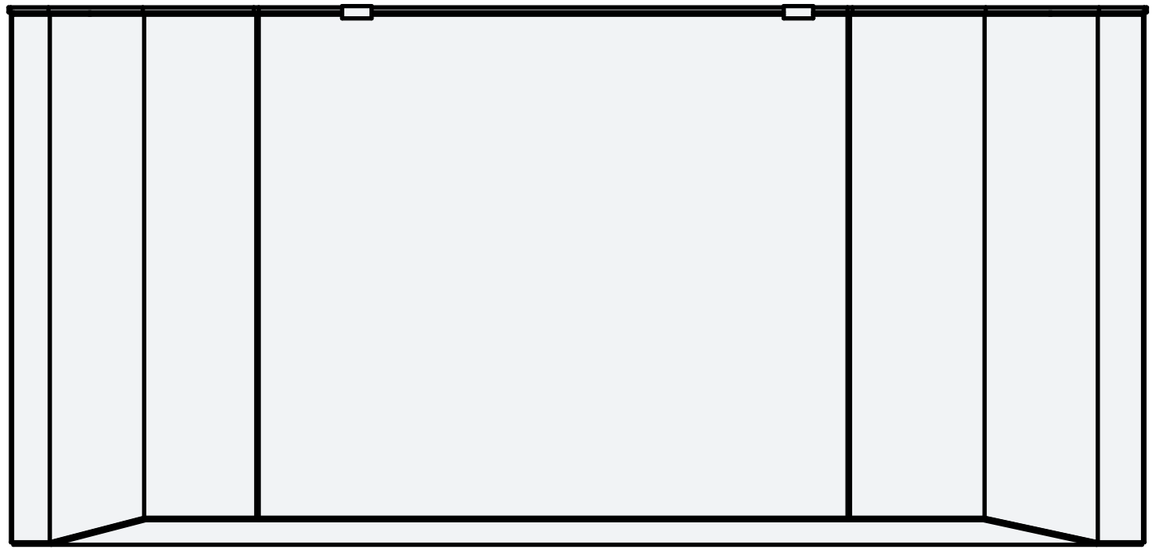


State S6: Theta=60 deg; doors open; downlock released; pitch and stroke are parent expressions
Projected PartIDs include leg, bogie, wheels, stays, fixed-length actuators and uplock hardware
Motion view is a geometry snapshot; clearance sign-off uses the separate native clearance report
Kinematics source: RLG003_MechKinematics; pin/bore fits and door gaps are preliminary drawing values

Stowed assembly and preliminary BOM summary

State S9 | Theta=90 deg | Stroke=0 mm | Beta=0 deg | DoorAngle=0 deg | Downlock=0.0 Uplock=1.0





State S9: Theta=90 deg; Stroke=0 mm; Beta=0 deg; Uplock=1; doors closed
BOM summary below is calculated from MassIncluded, MassOverrideKg and DensityKgM3 on saved solids
BOM solids: 595/670 mass-included; included mass 8010.06 kg (CAD properties)
Oleo/gear 67 parts; bogie/wheels 380 parts; mechanism 165 parts
Auxiliary 17 parts; reference/fluid 41 parts; total geometric mass 10711.6 kg
Mass and material assignments remain preliminary pending supplier allowables and final fastener BOM.